

Victor Villin

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PhD in machine learning. Expert in reinforcement learning, reward inference, and multi-agent systems. I design theoretically grounded learning algorithms and build scalable distributed training systems for autonomous agents. My experience spans mathematical modelling, deep reinforcement learning, and large-scale experimentation.

Publications

Workshop Papers and Preprints

Inferring Transferable Rewards via Active Inverse Reinforcement Learning *EWRL 2026 (Oral)*
Victor Villin, Till Freihaut, Andreas Schlaginhaufen, Giorgia Ramponi, Christos Dimitrakakis, Maryam Kamgarpour.

Published

Inference of Altruism and Intrinsic Rewards in Multi-Agent Systems *AAMAS 2026*
Victor Villin, Christos Dimitrakakis.

Fair Contracts in Principal-Agent Games with Heterogeneous Types *AAMAS 2026*
Jakub Tluczek, **Victor Villin**, Christos Dimitrakakis. Extended abstract.

A Minimax Approach to Ad Hoc Teamwork *AAMAS 2025*
Victor Villin, Thomas Kleine Büening, Christos Dimitrakakis.

Environment Design for Inverse Reinforcement Learning *ICML 2024 (Oral, Top 1%)*
Thomas Kleine Büening*, **Victor Villin***, Christos Dimitrakakis.

Effects of Different Optimization Formulations in Evolutionary Reinforcement Learning
on Diverse Behavior Generation *SSCI 2021*
Victor Villin, Naoki Masuyama, Yusuke Nojima.

* equal contribution.

Internships

Software Engineering Intern, Scalian, France *Summer 2019*

- Developed automation and software-validation tools for a 15-person engineering team modernising embedded systems for the French Leclerc tank.
- Streamlined repetitive verification workflows and improved software-upgrade validation by building practical GUI-based tools in Python, Shell, and C.

Selected Personal Projects

Polaris: Scalable Deep Multi-Agent Reinforcement Learning Framework *2024 – Present*
Built a modular distributed RL framework that enabled me to take new multi-agent algorithms from research ideas to large-scale experiments without relying on restrictive external training libraries. Implemented reusable infrastructure for asynchronous data collection, batched CPU/GPU inference, recurrent policies, experience processing, checkpointing, and experiment management, with base implementations of PPO, IMPALA, A3C, and V-MPO built on Ray and TensorFlow.  **Code**

Interactive RL Bot Tournament Platform for Super Smash Bros. Melee *2022 – Present*
Developed an online platform allowing users to create, customise, and deploy reinforcement learning

bots for Super Smash Bros. Melee. Bots learn by competing against each other in real time, with users controlling reward preferences and observing learning dynamics live.  [Code](#)  [seedsmash.ai](#)

Curriculum Reinforcement Learning for Pokemon Red 2024
Designed and implemented automated curriculum generation methods to solve Pokemon Red, a challenging, long-horizon, sparse-reward environment.  [Code](#)

Education

Ph.D. in Computer Science, University of Neuchâtel, Switzerland 2022 – Present
Expected graduation in October 2026.

Thesis: Environmental Interventions for Efficient Reward Inference and Cooperation in Multi-Agent Systems

Supervisor: Prof. Christos Dimitrakakis

M.Sc. in Computer Science, Osaka Metropolitan University, Japan 2021 – 2022
Double-degree programme with CY Tech.

Grade: A+

Thesis: Diverse Behaviour Generation in Evolutionary Reinforcement Learning

Supervisors: Prof. Yusuke Nojima and Dr. Naoki Masuyama

M.Sc. in Computer System Engineering, CY-Tech, France 2018 – 2022
Artificial Intelligence Specialisation.

Grade: A

Technical Skills

Research: reinforcement learning, machine learning, deep learning, preference elicitation, imitation learning, Bayesian inference, opponent modelling, game theory, active learning, social awareness, neuroscience, robustness

ML frameworks: TensorFlow, Ray, RLlib, PyTorch, JAX.

Programming: Python, C, C++, Java, Bash, JavaScript, Scala, Prolog.

Systems: Distributed learning, parallel simulation, Linux.

Awards

Swiss AI Small Grant 2026
LLMs as Proxies for Humans in Inverse Reinforcement Learning (Co-applicant).

Llamas'Partan Competition, CY-Tech, France 2021
Awarded for an outstanding personal project in reinforcement learning in video games.  [Code](#)

Teaching & Academic service

Teaching

Reinforcement Learning and Decision Making Under Uncertainty
Master course, Teaching Assistant, University of Neuchâtel.

Introduction to Computer Science
Bachelor course, Teaching Assistant, University of Neuchâtel.

Artificial Intelligence
Bachelor course, Teaching Assistant, University of Neuchâtel.

Reviewing

Reviewer for NeurIPS, AISTATS, and the European Workshop on Reinforcement Learning.

Additional Information

- **Track and Field:** Former national-level French sprinter; ranked ninth nationally in 2017.  **athle.fr**